

ABDUL RAHIMAN MIHAD

AI Engineer, LLM Agents and Real-Time Voice AI, Full-Stack and Cloud, MLOps

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PROFESSIONAL SUMMARY

AI Engineer building **Vendoor**, a B2B supplier discovery and tender platform for the UAE food and beverage market, owning the product end to end from four production LLM agents and WhatsApp and email automation to a FastAPI and PostgreSQL backend on AWS. Previously built **Alpha**, a full-duplex real-time AI voice interviewer at HireUp, lifting navigation accuracy from about 80% to 100%. Strong in Generative AI, agentic LLM systems, RAG, NLP, Computer Vision and MLOps. Microsoft Azure certified.

CORE COMPETENCIES

Artificial Intelligence (AI), Machine Learning (ML), Generative AI, Large Language Models (LLM), LLM Agents and Tool Use, Natural Language Processing (NLP), Computer Vision, Conversational AI, Voice AI, RAG, Vector Search, Prompt Engineering, LLM Fine-Tuning, Backend Architecture, Cloud Architecture (AWS), MLOps

WORK EXPERIENCE

Full-Stack AI Engineer, Vendoor

Jul 2026 to Present

B2B supplier discovery and tender platform for the UAE food and beverage market, UAE, early-stage startup

- **Built the platform end to end**, from the buyer-facing product and founder console to the AI pipeline, messaging and cloud infrastructure, taking it from MVP to production.
- **Developed four production LLM agents** with enforced structured output that turn buyer specs, spreadsheets and 5,000-SKU supplier catalogs into line items and parse free-text supplier quotes with per-line confidence.
- **Automated supplier outreach and quote collection** over the WhatsApp Cloud API and email, with HMAC-verified webhooks and atomic ingestion so retries never duplicate a quote and no message waits on a human.
- **Built the pricing normalisation engine and semantic catalog matching** with Voyage AI embeddings on pgvector, making every supplier price comparable at the buyer's quantity and matching aubergine to eggplant.
- **Migrated the backend** from Node.js and MySQL to Python 3.12, FastAPI and PostgreSQL on AWS (App Runner, ECS Fargate, RDS, S3, SQS), porting all 86 API routes under a CI-enforced contract with a zero-downtime cut-over.

AI/ML Engineer, HireUp

Jan 2026 to Jun 2026

Technology company headquartered in Japan (remote), Built Alpha, a real-time AI voice interviewer, [app.hireup.now](#)

- **Designed the full-duplex real-time voice pipeline** with WebSocket streaming, concurrent asyncio loops, Deepgram Flux STT, WebRTC VAD and streaming TTS, **cutting startup latency by about 1.5 seconds**.
- **Raised interview navigation accuracy from about 80% to 100%** by replacing non-deterministic LLM routing with a deterministic resolver, verified across live interviews and automated tests.
- **Significantly cut LLM operating cost** by redesigning the conversation turn loop to remove redundant model calls from every turn.
- **Built an automated evaluation harness** that replays full interviews in seconds, with MLflow and LangSmith tracing to catch regressions across releases.

KEY PROJECTS

Self-Corrective RAG System (LangGraph, AWS Bedrock, FastAPI, MLflow, Docker, MCP), [Live Demo](#)

- **Reduced hallucinations by 40% and reached 92% accuracy** with a self-correcting multi-agent pipeline (document grading, hallucination checks, query rewriting) on Llama 3.1 70B via AWS Bedrock, with sub-1.5s responses.

End-to-End Litter Detection System (YOLOv8, PyTorch, OpenCV, CVAT, DVC, MLflow, Streamlit), [Live Demo](#)

- **Improved small-object detection accuracy (mAP) by 15%** on 1,500+ hand-annotated images across 50+ tracked experiments, deployed as a live Streamlit app on a reproducible DVC and MLflow pipeline.

Character-Level Language Model from Scratch (PyTorch, NumPy, custom autograd), [GitHub](#)

- Implemented a character-level language model and a **custom automatic differentiation engine** from first principles to understand how modern LLMs train and generate text.

EDUCATION

Bachelor of Computer Applications (BCA), Yenepoya University, India

Aug 2022 to Nov 2025

Specialised in Machine Learning, AI and Robotics, Industry-partnered programme with TCS and Microsoft

CERTIFICATIONS

Microsoft	Azure AI Fundamentals (AI-900), Azure Fundamentals (AZ-900), Data Fundamentals (DP-900)
IBM (Coursera)	Advanced LLM Fine-Tuning , Fine-Tuning Transformers , AI Agents with RAG and LangChain , LLM Architecture and Data Prep , Language Models with Transformers , Foundational Models for NLP , EDA for Machine Learning
Kaggle	Advanced SQL, Intro to SQL

TECHNICAL SKILLS

AI / LLM	Tool Use and Structured Output, RAG, Embeddings (Voyage AI, pgvector), LangGraph, LangChain, Hugging Face, MCP, PEFT/LoRA, PyTorch, TensorFlow, NLP, Computer Vision
Real-Time AI	WebSockets, asyncio, Deepgram STT and TTS, WebRTC VAD, Web Audio API, Streaming Pipelines
Backend	Python 3.12, FastAPI, SQLAlchemy, PostgreSQL, pgvector, Alembic, Node.js, Express, MySQL, Redis
Frontend	JavaScript, HTML, CSS, Tailwind, Streamlit
DevOps	Docker, GitHub Actions (OIDC), CI/CD, pytest, ruff, MLflow, LangSmith, DVC, Model Monitoring
Cloud	AWS (App Runner, ECS Fargate, RDS, S3, SQS, Secrets Manager, Bedrock), Azure, GCP, BigQuery
Integrations	Meta WhatsApp Cloud API and Flows, Resend, Supabase Auth, pypdf, openpyxl, fpdf2
Languages	English (Fluent), Malayalam (Native), Hindi (Elementary), Arabic (Elementary)